

Prediction of wastewater treatment plant performance using artificial neural networks

Maged M. Hamed*, Mona G. Khalafallah, Ezzat A. Hassanien

Department of Public Works, Faculty of Engineering, Sanitary and Environmental Engineering Laboratory, Cairo University, Giza 12211, Egypt

Received 6 December 2002; received in revised form 10 July 2003; accepted 15 October 2003

Abstract

Artificial neural networks (ANN) models were developed to predict the performance of a wastewater treatment plant (WWTP) based on past information. The data used in this work were obtained from a major conventional treatment plant in the Greater Cairo district, Egypt, with an average flow rate of 1 million m³/day. Daily records of biochemical oxygen demand (BOD) and suspended solids (SS) concentrations through various stages of the treatment process over 10 months were obtained from the plant laboratory. Exploratory data analysis was used to detect relationships in the data and evaluate data dependence. Two ANN-based models for prediction of BOD and SS concentrations in plant effluent are presented. The appropriate architecture of the neural network models was determined through several steps of training and testing of the models. The ANN-based models were found to provide an efficient and a robust tool in predicting WWTP performance.

© 2003 Elsevier Ltd. All rights reserved.

Keywords: Neural networks; Waste water treatment; Model studies; Prediction; Optimization; Biochemical oxygen demand; Suspended solids

1. Introduction

The proper operation and control of wastewater treatment plants (WWTPs) is receiving increasing attention because of the rising concern about environmental issues. Improper operation of a WWTP may bring about serious environmental and public health problems, as its effluent to a receiving water body can cause or spread various diseases to human beings. Such issues are now thoroughly studied in Egypt, a country that has witnessed a surge of interest in environmental issues. This has culminated by the promulgation of Law 4 on the environment in 1994, which was the first comprehensive environmental law in Egypt. As a part of regulating air, water, and habitats, Egyptian discharge standards impose restrictions on the quality of WWTPs effluents, to conserve the receiving water bodies and, consequently, human health.

A better control of a WWTP can be achieved by developing a robust mathematical tool for predicting the plant performance based on past observations of certain key parameters. However, modeling a WWTP is a difficult task due to the complexity of the treatment processes. The complex physical, biological and chemical processes involved in wastewater treatment process exhibit non-linear behaviors which are difficult to describe by linear mathematical models.

This paper presents predictive models based on the concept of artificial neural networks (ANNs) (or simply neural networks), a widely used application of artificial intelligence that has shown quite a promise in a variety of applications in engineering, pattern recognition, and financial market analysis. The ANN-based models are applied to a major WWTP in Egypt. The developed models are shown to perform consistently well in the face of varying accuracy and size of input data. Using these models, the plant operators will be able to have an assessment of the expected plant effluent, given the quality of the waste stream at input locations.

* Corresponding author. Tel.: +2-012-7340619.

E-mail address: maged@alumni.rice.edu (M.M. Hamed).

2. Background

2.1. Basics of ANNs

The ANNs are mathematical modeling tools that are especially useful in the field of prediction and forecasting in complex settings. Historically, there were meant to operate through simulating, at a simplified level, the activity of the human brain. The ANN accomplishes this through a large number of highly interconnected processing elements (neurons), working in unison to solve specific problems, such as forecasting and pattern recognition. Each neuron is connected to certain of its neighbors with varying coefficients or weights, that represent the relative influence of the different neuron inputs to other neurons.

There are many kinds of ANNs. One of the most common types of neural networks is the feed-forward, where the information is transmitted in a forward direction only. This is the type employed in this work. A simple example of a neural network is shown in Fig. 1, consisting of one input layer, one hidden layer and one output layer, all of them connected with no feedback connections. The weighted sum of the inputs are transferred to the hidden neurons, where it is transformed using an activation function. The outputs of the hidden neurons, in turn, act as inputs to the output neuron where it undergoes another transformation. The output of a feed-forward neural network with one hidden layer and one output neural network is given by

$$\hat{Y}_t = f_o \left[\sum_{j=1}^{HN} WO_j * f_h \left(\sum_{i=1}^m WH_{ij} \cdot X_{it} + b_j \right) \right] + b_o \quad (1)$$

where WH_{ij} is the weight of the link between the i th input and the j th hidden neuron, m is the number of input neurons, WO_j is the weight of the link between the j th hidden neuron and the output neuron, f_h is the hidden neuron activation function, f_o is the output neuron

activation function, b_j is the bias of the j th hidden neurons, b_o is the bias of the output neuron, and HN is the number of hidden neurons. In this study, the tan-sigmoid activation function, $f_h(x) = 1/(1 + e^{-x})$, is used for the input and hidden layers, and the linear activation function, $f_o(x) = x$, is used as the output activation function.

The linear activation function for the output neuron was selected since it is appropriate for continuous-valued targets (Sarle, 2002), such as the biochemical oxygen demand (BOD) and suspended solids (SS) modeled in this work, especially for cases where the terms are believed to be additive. The tan-sigmoid activation functions for the input and hidden neurons are needed to introduce non-linearity into the network. Without non-linearity, hidden layers would not make nets more powerful than just plain perceptrons (which do not have any hidden units, just input and output units). Sigmoid activation functions are usually preferable to threshold activation functions. Networks with threshold functions are difficult to train because the error function is stepwise constant, hence the gradient either does not exist or is zero, making it impossible to use backpropagation or more efficient gradient-based training methods. With sigmoid function, a small change in the weights will usually produce a change in the outputs, which makes it possible to assess the changes in weights. With threshold function, a small change in the weights will often produce no change in the outputs (Sarle, 2002).

Network *training* is a process by which the connection weights and biases of the ANN are adapted through a continuous process of simulation by the environment in which the network is embedded. The primary goal of training is to minimize an error function by searching for a set of connection strengths and biases that causes the ANN to produce outputs that are equal or close to targets. In other words, training aims at estimating the parameters (WH_{ij} , WO_j , b_j , and b_o) by minimizing an error function, such as the mean square error (MSE) of the output values expressed as:

$$MSE = \sum_{t=1}^N (Y_t - \hat{Y}_t)^2 / N \quad (2)$$

where N is the number of data points. The minimization procedure relies on a numerical optimization of a non-linear objective function. A number of optimization routines can be used. In practice, the Levenberg–Marquardt routine often finds better optima for a variety of problems than do the other optimization methods (Sarle, 2002).

The learning process may be *supervised* or *unsupervised*. Supervised learning incorporates an external teacher, so that each output neuron is told what its desired response to input signals ought to be. During

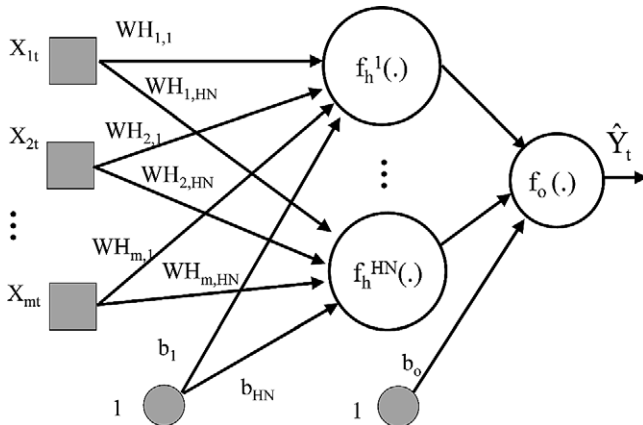


Fig. 1. Basic structure of an ANN model (after Awadallah and Rousselle, 1999).

the learning process, global information may be required. An important issue concerning supervised learning is the problem of error convergence, i.e. the minimization of error between the desired and computed unit values. The aim is to determine a set of weights which minimizes the error.

Unsupervised learning, on the other hand, uses no external teacher and is based upon only local information. In fact, for most varieties of unsupervised learning, the targets are the same as the inputs (Sarle, 1994). It is also referred to as self-organization, in the sense that it self-organizes data presented to the network and detects their emergent collective properties. In other words, unsupervised learning usually performs the same task as an auto-associative network, compressing the information from the inputs (Deco and Obradovic, 1996). Unsupervised learning is very useful for data visualization (Ripley, 1996). In this work, supervised learning was used, as explained in the later sections.

The architecture of a neural network consists of a description of the network's number of layers, the number of neurons in each layer, each layer's activation function, and how the layers connect to each other (Fausett, 1994). Learning rate parameter may also play an important role in the convergence of the network, depending on the application and the network architecture. The learning rate is used to increase the chance of avoiding the training process from being trapped in a local minimum instead of the global minimum. The network architecture and learning rate were determined in this work using a trial-and-error approach, as explained in the later discussion.

2.2. Examples of water and wastewater applications of ANNs

Recently, the ANNs have been increasingly applied in the area of environmental and water resources engineering. Govindaraju (2000) (ASCE Task Force) presents an extensive review of the application of ANNs in hydrology, where Maier and Dandy (2000) provide a review of the literature on the use of ANNs in water resources modeling. Following are the examples of some recent work done in the application of neural network to water and wastewater engineering problems. Neelakantan et al. (2001) developed a simple feed-forward backpropagation neural network to relate peak *Cryptosporidium* and *Giardia* concentrations with other biological, chemical and physical parameters in the Delaware River. Parameters significant to the identification of protozoa were identified by a stepwise elimination technique. The developed neural network model was successful in predicting the peaks of these pathogen concentrations. Neelakantan et al. (2002) then proceeded to identify the training algorithms that would maximize the accuracy of risk ranking for

Cryptosporidium and *Giardia* in case of relatively small datasets. They tested training algorithms based on gradient search with first- and second-order partial derivatives (variations of the steepest descent, conjugate gradient and Newton's) and genetic algorithms. Neural network training based on the genetic algorithms consistently provided better results compared to the other methods.

Djebbar and Narbaitz (2002) developed a neural network model for the analysis of the mass transfer coefficient in air stripping packed towers, used to remove volatile organic compounds. The model resulted in a considerable improvement over existing statistical approaches, and was able to simulate the sudden increase in overall mass transfer coefficient at high gas loading rates. It also simulated more realistically the effect of the packing depth and liquid flow. Holubar et al. (2002) applied neural networks based on the feed-forward backpropagation algorithm to model and control methane production in anaerobic digesters. The model was trained using data generated from four anaerobic continuous stirred tank reactors operating at steady state. The set of modeling parameters included gas composition, methane production rate, volatile fatty acid concentration, pH, redox potential, volatile SS and chemical oxygen demand of feed and effluent. The authors were able to predict gas production several steps in advance to predict and avoid shock-loads.

Chen et al. (2001) proposed a new approach for modeling industrial WWTPs. The approach employed a three-stage analysis that integrates fuzzy logic, genetic algorithms, and neural networks. The new approach was used for generating a representative state function, searching a set of multiobjective control strategies, and automatically tuning the fuzzy control rule base used for controlling a treatment plant. The developed model was successfully applied on a case study in Taiwan. Moreno-Alfonso and Redondo (2001) proposed an *intelligent* wastewater treatment concept, aided by two sets of neural networks, with the aim of controlling the plant in terms of previously selected parameters. The first neural network is used for *reasoned* (regular) control of the plant. The other neural network is termed *instinctive*, and is dedicated to the monitoring of the critical parameters. Its output determines whether a drastic decision, like stopping the process, is warranted. They contend that such a two-network system will prove useful in the management of WWTPs. Serodes et al. (2001) developed a decision support tool, *Chlorocast*, based on neural networks for modeling the chlorination process in the final disinfection phase of a water treatment system. They illustrated the power of the developed tool by applying it to forecast the residual chlorine in the drinking water tank and distribution system of the city of Sainte-Foy.

In so doing, the tool relied on representative historical data of the evolution of residual chlorine.

Zhang and Stanley (1999) studied the capacity of neural networks for process control in water treatment processes. Their proposed ANN-based model formed the major component of a software controller for coagulation, flocculation, and sedimentation processes. The model was found to consistently predict the optimum alum and powder activated carbon doses for different control actions. Tay and Zhang (1999) integrated fuzzy systems and neural networks in modeling the complex process of anaerobic biological treatment of wastewater. They illustrated the power of the technique in two case studies of upflow anaerobic sludge blanket and anaerobic fluidized bed reactor. The fuzzy-neural model simulated the system performance

well and provided satisfactory prediction results based on observed past information, although a disadvantage of the model was its high dependence on the quality of the training data.

Delgrange et al. (1998) applied neural networks to model the ultrafiltration membranes used in drinking water treatment. They experimented with a number of different network structures and architectures to compute the pressure at the end of filtration cycle and after the next backwash. The input parameters were the permeate flow rate, turbidity during the cycle and pressure measurements at the cycle start and at the end of the previous cycle. The networks were able to model, with good accuracy, the effect of both reversible and irreversible fouling on pressures.

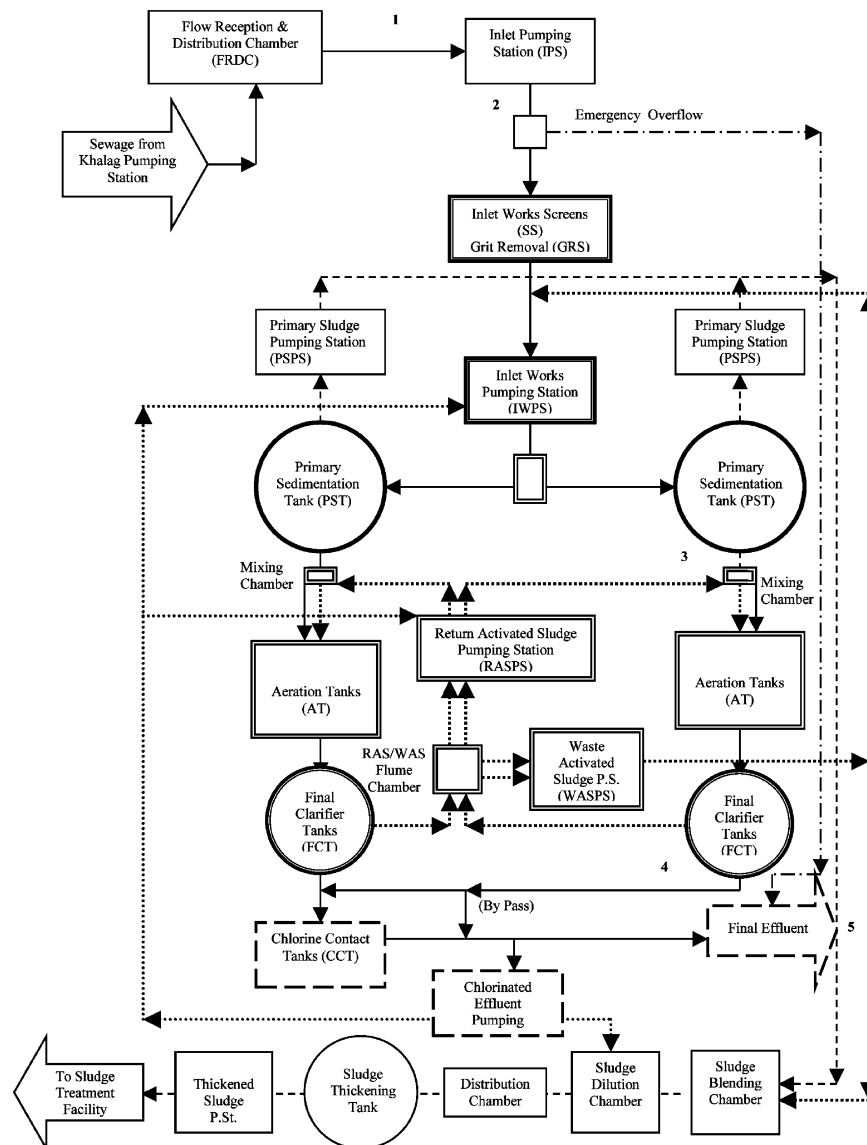


Fig. 2. Schematic of the wastewater processes at El-Gabal El-Asfar treatment plant, Greater Cairo area, Egypt. Note: numbers refer to sampling locations, and are cross referenced with Table 2 and Fig. 6.

3. Case study

The developed ANN models are applied to El-Gabal El-Asfar WWTP, which is the main WWTP on the east bank of the river Nile, Egypt, and is considered the largest WWTP in the Middle East. It is being constructed in three stages to serve an ultimate population of 12 millions. The existing first stage receives an average flow of 1 million m³/day and peak diurnal flow of 1.5 million m³/day. It is expected that the capacity of the plant will reach 3 million m³/day after completion of the three stages. The plant is of a conventional non-nitrifying activated sludge type and includes screening and grit removal, primary sedimentation, surface aeration, final clarification and chlorination facilities. Stage units are arranged in four modules (streams), each providing treatment for 0.25 million m³/day with common ancillary service buildings. The sludge digestion facilities are arranged into two modules with common mechanical dewatering, power generation and ancillary service buildings. A schematic of the plant processes is shown in Fig. 2.

Daily records over the period of 10 months were obtained from the treatment plant for concentrations of BOD and SS at different locations through the treatment process. Specifically, measurements are made at the forward reception and distribution chamber (FRDC), at input pumping station (IPS), following the primary sedimentation tanks (PST), and at the effluent stream from the final stage, which is composed of aeration tanks (AT) and final clarifier tanks (FCT). An example of the effluent BOD and SS in mg/l at the plant exit is shown in Figs. 3 and 4, respectively. Also shown in the figures are box plots of the data. Some of the daily records were missing, and at times only one of the two attributes are reported. This explains the disparity between the number of sample sequence numbers for the BOD and SS.

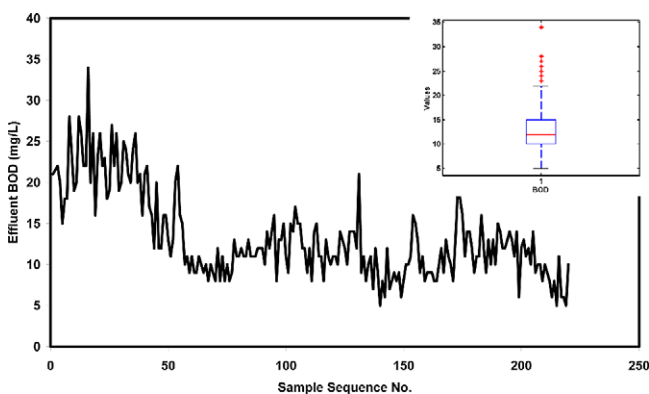


Fig. 3. Effluent BOD concentration at plant exit, including box plot of the data.

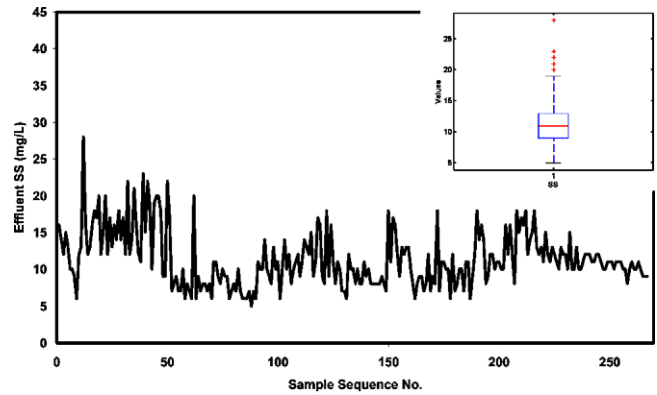


Fig. 4. Effluent SS concentration at plant exit, including box plot of the data.

4. Model development

A number of steps were carried out during the model development process. These are shown schematically in Fig. 5. The ANN model shares many of the attributes and steps relevant to other models of the non ANN-types. The analogy is shown in Table 1.

4.1. Data collection and preprocessing

The main objective of the data preprocessing is to determine the suitable locations for the data required for the modeling activities. Although a number of parameters may impact the plant performance (e.g.

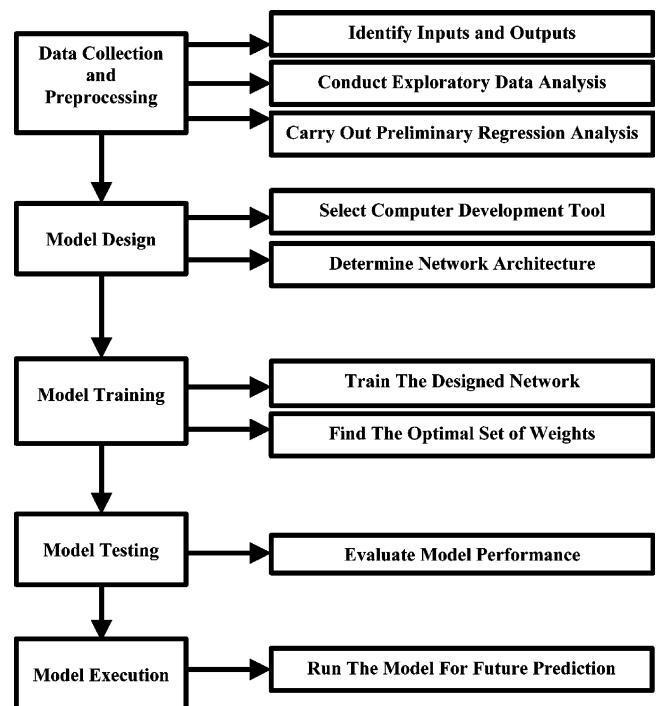


Fig. 5. Steps of the model development process.

Table 1
Analogy between (classic) mathematical and ANN modeling processes

Classic (non-ANN) mathematical models	ANN models
Conceptual design	Network architecture identification
Calibration	Model training
Validation	Model testing
Model prediction	Model execution

pH, temperature, chemical dosages), the only data reported by the plant operators were BOD and SS. Therefore, this work only relied on these two parameters. In spite of the limitation this issue may present, the two parameters are considered good indicators of the plant performance as reported in the literature (c.f. Droste, 1997).

Exploratory data analysis was used to identify the type and strength of relation between inputs and outputs. Scatter diagrams were constructed for several cases of inputs and outputs, and input/output sampling locations of the measured data. For each case, a linear least square regression analysis was conducted and the coefficient of determination (R^2) was computed to determine how a variation in the input parameter(s) could explain variations in the corresponding output. A list of the different cases considered is shown in Table 2. The position numbers referenced in the table correspond to specific locations along the treatment train, as shown in Fig. 6. Cases 17 and 18 were selected for this paper, as they presented greater values of R^2 than other cases. The results of the regression analysis for cases 17 and 18 are shown in Figs. 7 and 8. Therefore, two models were developed: one that uses BOD

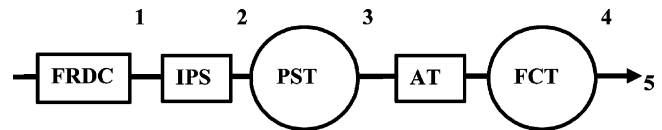


Fig. 6. Sampling locations along the treatment train.

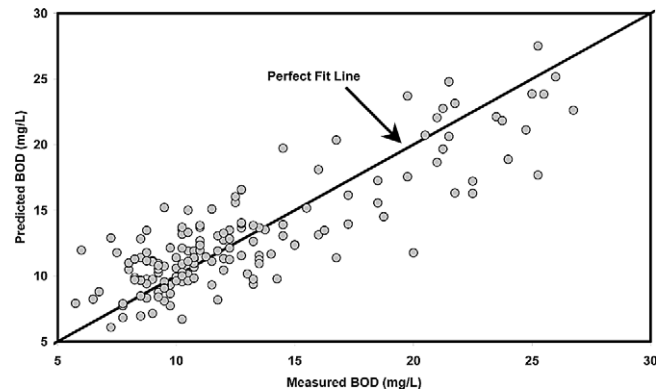


Fig. 7. Measured versus regression-based predicted BOD for case 17.

and SS concentrations downstream of PST as inputs to predict FCT effluent BOD concentration (case 17) and the other uses BOD and SS concentrations downstream of PST to predict SS concentration in the final clarifier effluent (case 18).

4.2. Model design and network training

The model architecture, training method, and training rates were determined using a trial-and-error approach.

Table 2
Cases of inputs and outputs versus R^2

Case no.	Input parameter	Position	Output parameter	Position	R^2
1	BOD from FRDC	1	BOD avg. ^a from PST	3	0.269
2	SS from FRDC	1	SS avg. from PST	3	0.210
3	BOD from IPS	2	BOD avg. from PST	3	0.530
4	SS from IPS	2	SS avg. from PST	3	0.353
5	BOD avg. from PST	3	BOD avg. from FCT	4	0.641
6	SS avg. from PST	3	SS avg. from FCT	4	0.536
7	BOD avg. from PST	3	BOD final effluent	5	0.516
8	SS avg. from PST	3	SS final effluent	5	0.251
9	BOD from FRDC	1	BOD avg. from FCT	4	0.290
10	SS from FRDC	1	SS avg. from FCT	4	0.134
11	BOD from IPS	2	BOD avg. from FCT	4	0.228
12	SS from IPS	2	SS avg. streams effluent	4	0.235
13	BOD from FRDC	1	BOD final effluent	5	0.200
14	SS from FRDC	1	SS final effluent	5	0.097
15	BOD from IPS	2	BOD final effluent	5	0.128
16	SS from IPS	2	SS final effluent	5	0.151
17	BOD and SS avg. from PST	3	BOD avg. from FCT	4	0.665
18	BOD and SS avg. from PST	3	SS avg. from FCT	4	0.542

FRDC, flow reception and distribution chamber; IPS, inlet pumping station.

PST, preliminary sedimentation tank; FCT, final clarifier tanks.

^a avg. refers to the average value calculated from all four plant streams (A, B, C and D).

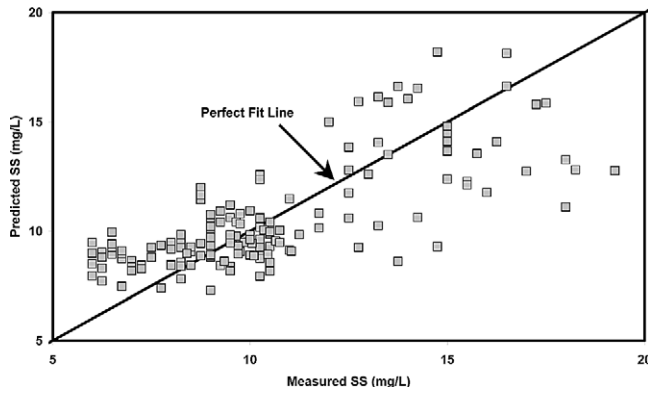


Fig. 8. Measured versus regression-based predicted SS for case 18.

The pair of BOD and SS data sets (153 sets for each of case 17 and 18) were divided into four groups each containing two subsets of data, namely *training* and *testing* data. Groups 1–4 used 60%, 70%, 80% and 90% of the data for training, respectively. This corresponds to training phase data sets containing 92, 108, 123, and 138 points. For example, the figure 92 corresponds to 60% of the complete data set of 153 values. The remaining data points were used for testing phase. So, for the case where 92 data points were used for training, the remaining 61 data points were used for testing purposes.

To reach the suitable network architecture, several trials for each group have been conducted until the suitable learning rate, number of hidden layers and number of neurons per each hidden layer were reached. The suitable architecture is the one which produced the minimal error term in both training and testing data.

In this work, a supervised training, backpropagation algorithm is selected. The backpropagation algorithm minimizes the MSE between the observed and the predicted output in the output layer, through two phases. In the forward phase, the external input information signals at the input neurons are propagated forward to compute the output information signal at the output neuron. In the backward phase, modifications to the connection strengths are made based on the basis of the difference in the predicted and observed information signals at the output neuron.

The performance of each network model is evaluated by computing the mean absolute percentage error (MAPE) and MSE for each trial conducted in search of the suitable architecture and for each of the four groups. The structure that resulted in minimum errors was the one selected. The suitable architecture for BOD prediction was determined to consist of an input layer with two neurons (for input BOD and SS), a

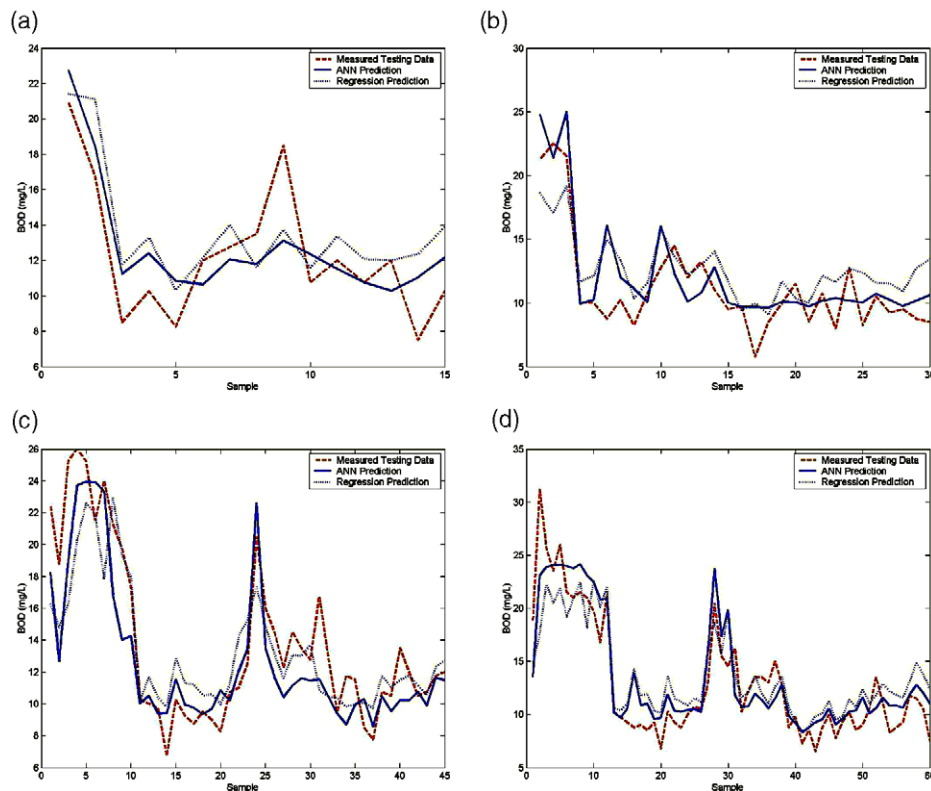


Fig. 9. Comparison of measured, ANN-modeled, and regression-modeled BOD concentrations for (a) Group 1, (b) Group 2, (c) Group 3, and (d) Group 4.

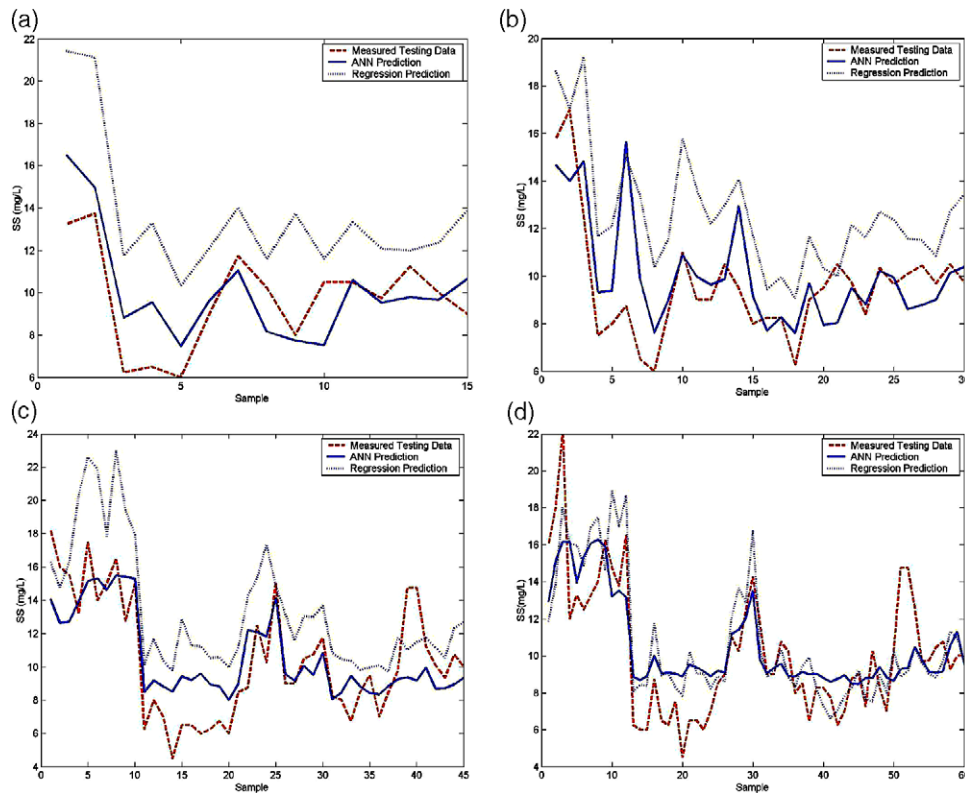


Fig. 10. Comparison of measured, ANN-modeled, and regression-modeled SS concentrations for (a) Group 1, (b) Group 2, (c) Group 3, and (d) Group 4.

hidden layer with three neurons and an output layer with one neuron (for predicted BOD).

The suitable network architecture for SS prediction, on the other hand, was determined to consist of an input layer with two neurons (for input BOD and SS), a hidden layer with two neurons and an output layer with one neuron (for predicted SS). To reach this finding, different network structure/training rate combinations were evaluated. Network structures with one hidden layer containing 2, 4, 6, 8, 10, 12, and 14 neurons were evaluated, each for training rates of 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1, 1.5, 2, 4. This makes the total number of combinations equal to seven structures times 10 learning rates, totaling 70 combinations. A number of training methods were used, including the Fletcher–Reeves conjugate gradient, BFGS quasi-Newton, and the Levenberg–Marquardt algorithms. The latter algorithm proved to be the fastest and most robust. A suitable learning rate was determined to be 1.0 for both models. One hidden layer was proved to be a universal approximator (Hornik et al., 1989). Both networks preserve the parsimony principle to improve the generalization capacity of the ANNs. Parsimony principle states that the use of simpler models, with fewer number of parameters, is usually preferable to more complicated ones with more parameters, whenever feasible (Lawrence et al., 1996).

During the training phase, the MSE ranged from 4.9 to 5.4, and the MSE for the testing phase ranged from 4.1 to 6.1. The mean absolute percent error ranged from around 13% to 17% for all groups. The narrow band of error measures throughout the groups, and for the two modeled parameters (namely, BOD and SS), is an indication of the ANN's robustness. In essence, the ANN managed to escape being trapped in noise present in the data. The robustness of the ANN models are also shown in the superiority of the ANN with respect to the regression models for all data division groups 1–4 (Figs. 9 and 10).

4.3. Model testing

Actual values of BOD and SS in the testing datasets are compared to predicted values by the neural network models, to evaluate the models performance. Furthermore, those were compared to the regression-modeled values. Figs. 9 and 10 show these comparisons for groups 1–4, for BOD and SS outputs. Visual inspection indicates that the ANN models resulted in a good fit for the measured BOD and SS data.

The quality of match between the ANN-modeled and measured concentrations was determined by regression analysis. Figs. 11 and 12 show the measured versus ANN-modeled concentrations for the testing data

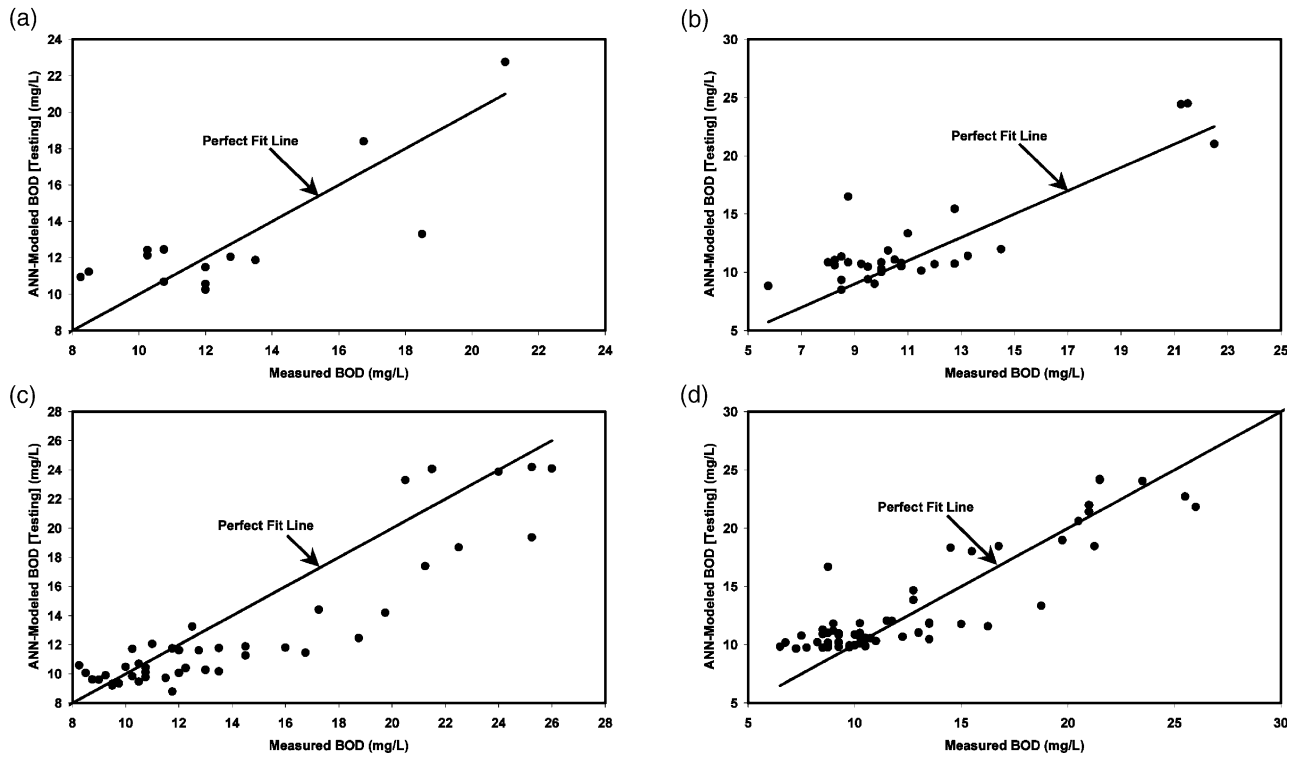


Fig. 11. Regression analysis of measured versus ANN-modeled BOD concentrations for testing data for (a) Group 1, (b) Group 2, (c) Group 3, and (d) Group 4.

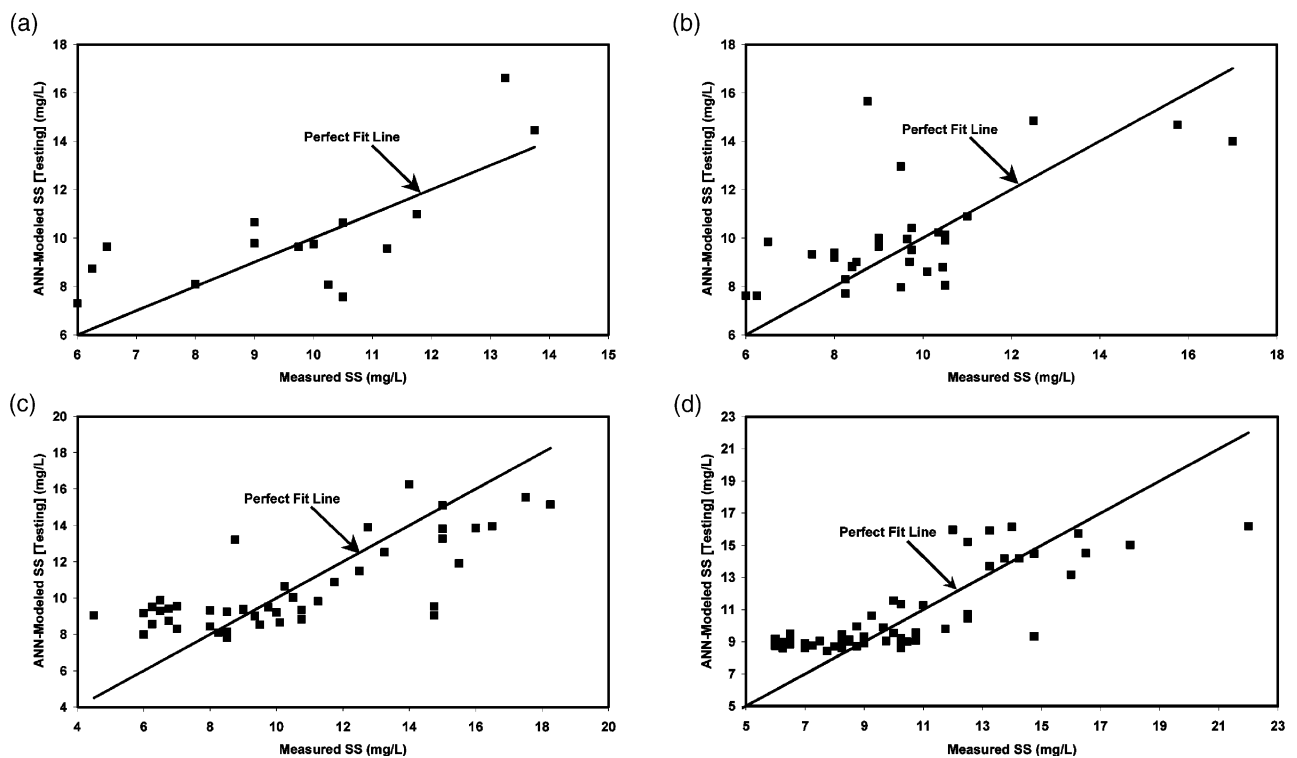


Fig. 12. Regression analysis of measured versus ANN-modeled SS concentrations for testing data for (a) Group 1, (b) Group 2, (c) Group 3, and (d) Group 4.

sets for Groups 1–4, for BOD and SS, respectively. R^2 values ranged from 0.63 to 0.81 for BOD, and from 0.45 to 0.65 for SS and for all cases shown. The low R^2 results obtained for the SS are likely the result of data noise.

5. Summary and conclusions

In this paper, two models based on ANNs were developed to predict the effluent concentrations of BOD and SS for a major WWTP in Cairo. The developed models were trained and tested on daily sets of BOD and SS measurements over a period of 10 months. The neural network models provided good estimates for the BOD and SS data sets, which cover a range of data for training and testing purposes.

The ANN models provided a robust tool for prediction in that the prediction error varied slightly, and smoothly, over the range of data sizes used in training and testing. The limitation in data, however, should be highlighted. If more data were collected, if the data were less noisy, and if additional parameters were measured (e.g. pH, temperature, etc.), this would have resulted in an improved predictive capability of the network. Nevertheless, the ANN is a tool that is worth consideration in the prediction of WWTPs.

Acknowledgements

The authors acknowledge the assistance provided by the *Cairo Wastewater Organization for Execution of the Greater Cairo Wastewater Project* in making available the data collected at El-Gabal El-Asfar WWTP required to carry out this work.

References

- Awadallah, A.G., Rousselle, J., 1999. Forecasting the Nile flood using sea surface temperatures as inputs: a comparison between transfer function with noise and neural networks. In: Proceedings of the 19th Annual American Geophysical Union Hydrology Days, Fort Collins, Colorado.
- Chen, W.C., Chang, N.B., Shieh, W.K., 2001. Advanced hybrid fuzzy-neural controller for industrial wastewater treatment. *Journal of Environmental Engineering* 127 (11), 1048–1059.
- Deco, G., Obradovic, D., 1996. *An Information—Theoretic Approach to Neural Computing*. Springer-Verlag, New York.
- Delgrange, V.N., Cabassud, N., Cabassud, M., Durand-Bourlier, L., Laine, J.M., 1998. Neural networks for prediction of ultrafiltration transmembrane pressure: application to drinking water production. *Journal of Membrane Science* 150, 111–123.
- Djebbar, Y., Narbaitz, R.M., 2002. Neural network prediction of air stripping KLa. *Journal of Environmental Engineering* 128 (5), 451–460.
- Droste, R.L., 1997. *Theory and Practice of Water and Wastewater Treatment*. John Wiley and Sons, Inc.
- Fausett, L., 1994. *Fundamentals of Neural Networks*. Prentice Hall, Englewood Cliffs, NJ.
- Govindaraju, R.S., 2000. Artificial neural networks in hydrology. II: hydrologic applications, ASCE task committee on application of artificial neural networks in hydrology. *Journal of Hydrologic Engineering* 5 (2), 124–137.
- Holubar, P., Zani, L., Hager, M., Froschl, W., Radak, Z., Braun, R., 2002. Advanced controlling of anaerobic digestion by means of hierarchical neural networks. *Water Research* 36, 2582–2588.
- Hornik, K., Stinchcombe, M., White, H., 1989. Multilayer feedforward networks are universal approximators. *Neural Networks* 2, 359–366.
- Lawrence, S., Giles, C.L., Tsoi, A.C., 1996. What size neural network gives optimal generalization? Convergence properties of back-propagation, Technical Report UMIACS-TR-96-22 and CS-TR-3617. Institute for Advanced Computer Studies, University of Maryland, College Park, MD 20742.
- Maier, H.R., Dandy, G.C., 2000. Neural networks for the prediction and forecasting of water resources variables: a review of modeling issues and applications. *Water Resources Research* 15 (1), 101–124.
- Moreno-Alfonso, N., Redondo, C.F., 2001. Intelligent waste-water treatment with neural-networks. *Water Policy* 3, 267–271.
- Neelakantan, T.R., Brion, G.M., Lingireddy, S., 2001. Neural network modelling of Cryptosporidium and Giardia concentrations in the Delaware River, USA. *Water Science and Technology* 43 (12), 125–132.
- Neelakantan, T.R., Lingireddy, S., Brion, G.M., 2002. Effectiveness of different artificial neural network training algorithms in predicting protozoa risks in surface waters. *Journal of Environmental Engineering* 128 (6), 533–542.
- Ripley, B.D., 1996. *Pattern Recognition and Neural Networks*. Cambridge University Press, Cambridge, UK.
- Sarle, W.S., 1994. Neural networks and statistical models. In: Proceedings of the 19th Annual SAS Users Group International Conference, Cary, NC, SAS Institute Inc, pp. 1538–1550.
- Sarle, W.S., 2002. “Neural Network FAQ, part 1 of 7: Introduction”, periodic posting to the Usenet newsgroup comp.ai.neural-nets. Available at: <ftp://ftp.sas.com/pub/neural/FAQ.html>.
- Serodes, J.B., Rodriguez, M.J., Ponton, A., 2001. Chlorcast: a methodology for developing decision-making tools for chlorine disinfection control. *Environmental Modelling and Software* 16, 53–62.
- Tay, J.H., Zhang, X., 1999. Neural fuzzy modeling of anaerobic biological wastewater treatment systems. *Journal of Environmental Engineering* 125 (12), 1149–1159.
- Zhang, Q., Stanley, S.J., 1999. Real-time water treatment process control with artificial neural networks. *Journal of Environmental Engineering* 125 (2), 153–160.